

COMP3013 Database Management System

Course Project Requirements 2026 Spring

Introduction

In this project, you are expected to model, design, and implement a real-world web application related to data analysis and database management. For example, a simple music management system, a university course arrangement system, a library book management system, an e-business inventory management system, a train ticket booking system, etc. (you are encouraged to propose your own application; innovative ideas will receive bonus points). You should do the implementation using Python, Django framework (or other web development frameworks) and MySQL DBMS. Your web application should provide users with reasonable operations based on your application and problem.

This is a **group-based** project. Each group consist of **4 students** and you should choose your group via the grouping link on iSpace.

Workflow

The project workflow goes through several stages described below.

1. **Problem definition.** Each group will need to select a topic and formulate the problem, e.g., music management, e-business inventory management or ticket booking, etc.
2. **Data collection.** Each group will need to collect the data using the web scraping techniques or download existing datasets. You can also generate synthetic datasets if necessary.
3. **Wrangle, prepare, cleanse the data.** The collected data may contain missing and noisy entries or inconsistent schemas. Therefore, you need to perform data preprocessing and data cleaning.
4. **Database design.** Design the database using the ER Model by creating an ER diagram and later on converting it into relations in MySQL DBMS.
5. **Web development.** Design and implement the website that provides a user interface to manage the database, including but not limited to query the database, insert/update/delete some records in the database relations.

Requirements

Front End Requirements

- Your website needs at least 8 pages, each page needs to include navigation bars, logo, footers, etc.
- Your website should allow user registration and login. At least two user types should be offered. For example, suppose you are building a bookstore website, your user types could be *administrator*, *registered user* and *anonymous visitor*.
- Your website needs to include at least four features. The more features you have, the higher score you will earn. Feature is a workflow that can allow the user to perform a complete task. Take the example of a bookstore, the features could be:
 - A user searches book.
 - An admin adds new books.
 - An admin views purchasing history and statistical/analysis results.
 - ...

Back End Requirements

- Your ER diagram should have at least 8 entities and 6 relationships.
- On average, each table must have no less than 5,000 records. There are at least two tables consisting of more than 50,000 records. (Note: Not all the records have to be real but they should be realistic. You may generate records using a program.)
- Your system takes less than 1 second to insert a record to any table.
- Your system takes less than 2 seconds to delete or search a record.
- You may have to create an index to accelerate deletion or search.
- The logical design of the database must follow the normal forms.

Presentation Requirements

- 10 minutes (pre) + 2 minutes (QA) for each group.
- Make some good slides.

Report Requirements

- Project description. Briefly introduce the purpose of this project. You should define which real-life problem you are solving, address the difficulty of the problem, give the abstraction of the problem, and major goal of the project.
- Introduce the dataset, data collection and data preprocessing approaches in your project.
- To design the database, you may need to make some assumptions for modeling because real-life problems sometimes do not provide enough information. But, all your assumptions have to be realistic.
- Your report must include your final ER diagram and a brief description of each entity and relationship set.
- You need to give all functional dependencies and schemas.
- If your schemas are in the normal forms, explain why. If not, you need to decompose them and show the steps in detail.
- You need to describe the primary keys.
- List all front-end and back-end functions that you have implemented.
- List the SQL codes and explanations of main functions in your project.
- **Workload and contribution of each team member.**

Bonus (Maximum 10%)

- Use triggers to implement constraints other than *not null*, *primary key*, *unique*, or *referential*. For example, the table *student* has an attribute *credit* showing the total credits earned by each student. Then, if a student finishes a course of 3 credits, his/her *credit* will be increased by 3 automatically.
- Other outstanding or innovative features in addition to the implementation of correct and complete fundamental functions.

Presentation & Report Tips

- The presentation should follow the top-down procedure. Please start your presentation from the problem definition and assume that audiences know nothing about your project.

- Describe your database design on a high level, ER diagram for example.
- Highlight the critical points, constraints and triggers for example.
- Prepare some good slides.
- Do **NOT** make your presentation as function demonstration.
- Time is limited. You cannot show all details in a short presentation. Attract people's attention by some shining gold.

Grading

- Report 30%
- Presentation 10%
- Website Implementation 20%
- Database Implementation 40%
- Bonus 10% (maximum)

Submission

May 11th - May 29th: Project presentation and **final submission**. The submission includes:

- The report of the whole project.
- All code for your application, including the web scraping code, data preprocessing and analysis code, database code (.sql) and web development code.
- Any auxiliary files needed by your program to create the database and application.

One group only submits **one copy** to iSpace **by the team leader**, rename your final submission package as “COMP3013_Group###.zip”.